

Practice Problems Set 1: Answers/solutions

Problem 1

Multiplying a number by j has the effect of rotating it anticlockwise by 90 degrees.

Problem 2:

Solution: Using Cartesian form:

$$\frac{1-j}{1+j} = \frac{(1-j)^2}{(1+j)(1-j)} = \frac{-2j}{2} = -j$$

Other way:

$$\begin{aligned} 1-j &= \sqrt{2}e^{-j\pi/4} \\ 1+j &= \sqrt{2}e^{j\pi/4} \\ \frac{1-j}{1+j} &= \frac{\sqrt{2}}{\sqrt{2}} \cdot e^{(-\pi/4 - \pi/4)j} \\ &= e^{-j\pi/2} = -j \end{aligned}$$

Problem 3:

Solution:

$$\begin{aligned} a) \quad (1-j)^4 &= 1 + 4(-j) + 6(-j)^2 + 4(-j)^3 + (-j)^4 \\ &= 1 - 6 + 1 + j(-4 + 4) = -4 \end{aligned}$$

By DeMoivre:

$$\begin{aligned} 1-j &= \sqrt{2}e^{-j\pi/4} \\ (1-j)^4 &= (\sqrt{2})^4 e^{-j\pi} = 4 \cdot (-1) = -4 \end{aligned}$$

$$\begin{aligned} b) \quad (1+j\sqrt{3})^3 &= 1 + 3(j\sqrt{3}) + 3(j\sqrt{3})^2 + (j\sqrt{3})^3 \\ &= 1 + 3j\sqrt{3} + 3 \cdot (-3) + j^3 3\sqrt{3} \\ &= -8 + j(3\sqrt{3} - 3\sqrt{3}) = -8 \end{aligned}$$

By DeMoivre:

$$\begin{aligned} 1+j\sqrt{3} &= 2e^{j\pi/3} \\ (1+j\sqrt{3})^3 &= 8e^{j\pi} = -8 \end{aligned}$$

Problem 4:

Solution: The rectangular expression gives the coordinates for the little pictures. Any angle may be altered by adding a multiple of 2π .

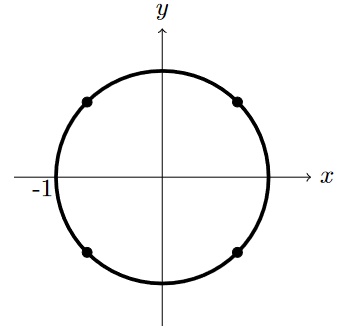
$1-j$	$\sqrt{2}, -\pi/4$	$\sqrt{2}e^{-j\pi/4}$
$\sqrt{3}+j$	$2, \pi/6$	$2e^{j\pi/6}$
$(-1-j)/\sqrt{2}$	$1, 5\pi/4$	$e^{5j\pi/4}$
$(1+\sqrt{3}j)/2$	$1, \pi/3$	$e^{j\pi/3}$
$(-1+j)/\sqrt{2}$	$1, 3\pi/4$	$e^{3j\pi/4}$

Problem 5: Find the complex roots of the equation: $x^4 + 16 = 0$

Solution: $\sqrt[4]{-16} = 2 \cdot \sqrt[4]{-1}$

The 4th roots of -1 are $e^{j(\pi/4 + n\pi/2)} = \frac{\pm 1 \pm j}{\sqrt{2}}$. They are shown in the figure

Thus, $\sqrt{2}(\pm 1 \pm j)$ are the roots of $x^4 + 16 = 0$.



Problem 6: Show that

a) $\cos \theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$

b) $\sin \theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$e^{-j\theta} = \cos \theta - j \sin \theta \quad \text{since} \quad \begin{matrix} \cos(-\theta) = \cos \theta \\ \sin(-\theta) = -\sin \theta \end{matrix}$$

a) Adding: $e^{j\theta} + e^{-j\theta} = 2 \cos \theta$
 $\therefore \cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$

b) Subtracting: $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$
 $\therefore \sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$

Problem 7:

a) Find non-negative real numbers A, ω , and ϕ such that $\operatorname{Re}\left(\frac{je^{2jt}}{1+j}\right) = A \cos(\omega t + \phi)$

b) Express the cube roots of $8j$ in the form $a + bj$

Solution

a) Express all complex numbers in exponential form:

$$\frac{je^{2jt}}{1+j} = \frac{e^{j\pi/2} e^{2jt}}{\sqrt{2} e^{j\pi/4}} = \frac{1}{\sqrt{2}} e^{j(2t + \pi/2 - \pi/4)} = \frac{1}{\sqrt{2}} e^{j(2t + \pi/4)}$$

The real part is

$$\operatorname{Re}\left(\frac{je^{2jt}}{1+j}\right) = \frac{1}{\sqrt{2}} \cos\left(2t + \frac{\pi}{4}\right)$$

b)

The polar form of $8j$ is $8e^{j\pi/2}$. Its three cubic roots are

$$\begin{aligned} 2e^{j\pi/6} &= 2 \cos \frac{\pi}{6} + 2j \sin \frac{\pi}{6} = \sqrt{3} + j, \\ 2e^{j(\pi/6 + 2\pi/3)} &= 2 \cos \frac{5\pi}{6} + 2j \sin \frac{5\pi}{6} = -\sqrt{3} + j, \\ 2e^{j(\pi/6 + 4\pi/3)} &= 2e^{3j\pi/2} = -2j. \end{aligned}$$